

# WEEK 1 – AWS BASICS

## Practice Set Report

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**Program:** Skill-Nexis AWS Internship

**Week:** 1

This report documents the Week 1 AWS Basics practice set, covering cloud service models, IAM, EC2, AWS Regions and Availability Zones, and key AWS services.

# 1. IaaS, PaaS and SaaS

Cloud computing service models define how much of the underlying infrastructure is managed by the cloud provider and how much responsibility remains with the user.

| Model | Meaning                     | User manages                             | Example               |
|-------|-----------------------------|------------------------------------------|-----------------------|
| IaaS  | Infrastructure as a Service | OS, applications, data and configuration | Amazon EC2            |
| PaaS  | Platform as a Service       | Application code and data                | AWS Elastic Beanstalk |
| SaaS  | Software as a Service       | Mainly application usage and settings    | Gmail / Microsoft 365 |

**Summary:** IaaS provides infrastructure-level control, PaaS provides a managed application platform, and SaaS provides ready-to-use software.

# 2. Create and Configure an IAM User

An IAM user was created in the AWS account for identity and access management. The AWS Management Console confirms successful creation of the user **Aditya-SkillNexis**.

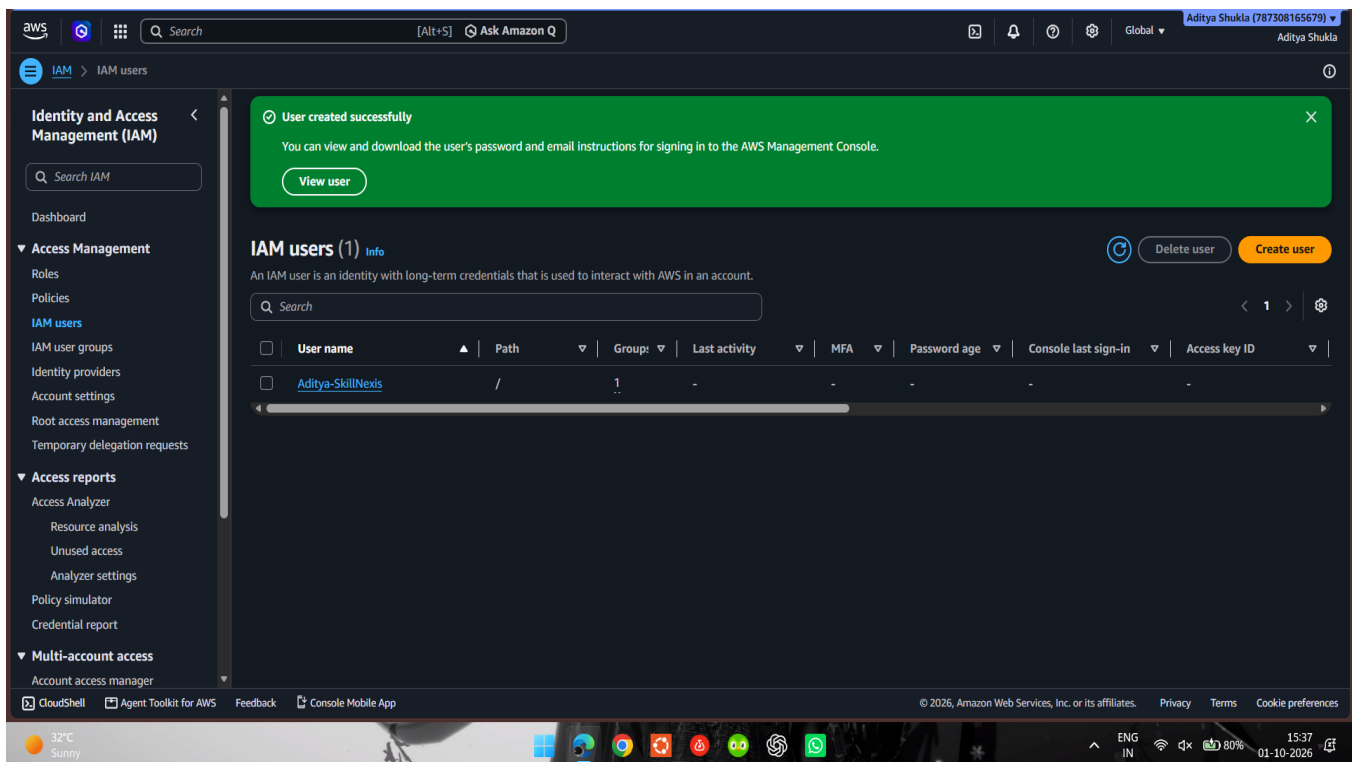


Figure 1. IAM user successfully created in AWS IAM.

### 3. Launch an EC2 Instance

An Amazon EC2 virtual machine was launched using Amazon Linux 2023. The instance was configured as a t3.micro instance and successfully reached the **Running** state.

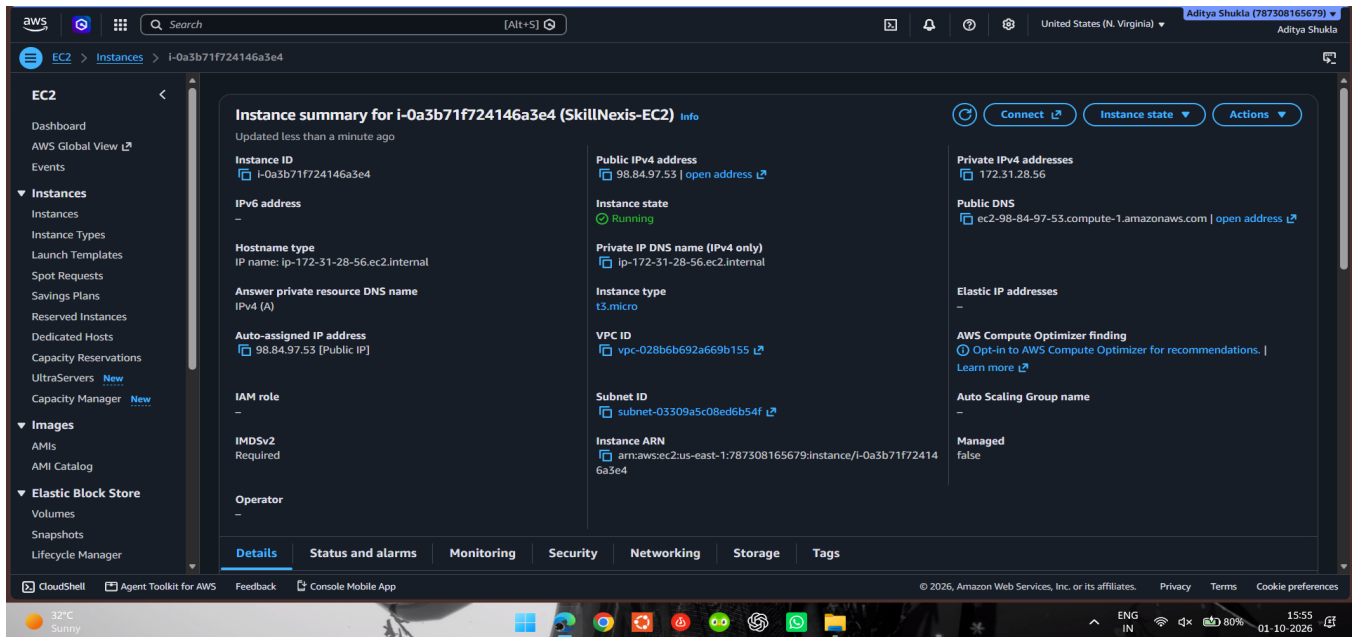


Figure 2. Running EC2 instance and its resource details.

## 4. AWS Regions and Availability Zones

An AWS Region is a geographic area containing multiple isolated Availability Zones. Availability Zones are separate infrastructure locations within a Region that support availability and fault isolation. The AWS Global View was explored to examine Regions, Region codes, and Availability Zone information.

The screenshot displays the AWS Global View interface, specifically the 'Regions and Zones' section. The page title is 'Regions and Zones' with an 'Info' icon. Below the title, a description states: 'AWS Regions are geographic areas containing multiple Availability Zones (isolated data centers), while Local Zones and Wavelength Zones bring AWS services closer to end-users and 5G networks respectively.' A search bar is present with the text 'Search Regions and Zones'. There are two buttons: 'List view' and 'Map view'. The 'Regions' tab is selected, showing a list of 1/34 regions. The list is updated 1 minute ago. The regions are listed in a table with columns: Region, Status, Region code, Geography, Availability Zones, Local Zones, and Wavelength Zones. The first region, Asia Pacific (Hyderabad), is selected and highlighted. The table shows the following data:

| Region                   | Status             | Region code    | Geography            | Availability ... | Local Zones | Wavelength Zones |
|--------------------------|--------------------|----------------|----------------------|------------------|-------------|------------------|
| Asia Pacific (Hyderabad) | Enabled            | ap-south-2     | Asia Pacific (Hyd... | 3                | -           | -                |
| Asia Pacific (Mumbai)    | Enabled by default | ap-south-1     | Asia Pacific (Mu...  | 3                | 2           | -                |
| Asia Pacific (Osaka)     | Enabled by default | ap-northeast-3 | Asia Pacific (Osa... | 3                | -           | -                |
| Asia Pacific (Seoul)     | Enabled by default | ap-northeast-2 | Asia Pacific (Seo... | 4                | -           | -                |
| Asia Pacific (Singapore) | Enabled by default | ap-southeast-1 | Asia Pacific (Sin... | 3                | 2           | -                |
| Asia Pacific (Sydney)    | Enabled by default | ap-southeast-2 | Asia Pacific (Syd... | 3                | 1           | -                |
| Asia Pacific (Tokyo)     | Enabled by default | ap-northeast-1 | Asia Pacific (Tok... | 3                | -           | 2                |
| Canada (Central)         | Enabled by default | ca-central-1   | Canada (Central)     | 3                | -           | 1                |
| Europe (Frankfurt)       | Enabled by default | eu-central-1   | Europe (Frankfurt)   | 3                | 4           | -                |
| Europe (Ireland)         | Enabled by default | eu-west-1      | Europe (Ireland)     | 3                | -           | -                |
| Europe (London)          | Enabled by default | eu-west-2      | Europe (London)      | 4                | -           | 1                |
| Europe (Paris)           | Enabled by default | eu-west-3      | Europe (Paris)       | 3                | -           | 2                |

Figure 3. AWS Global View showing Regions and Availability Zone information.

## 5. Summary of Key AWS Services

### Amazon EC2

Amazon Elastic Compute Cloud (EC2) provides resizable compute capacity in the AWS Cloud. It is commonly used to run virtual servers for web applications, backend services, development environments, and other workloads.

### Amazon S3

Amazon Simple Storage Service (S3) is an object storage service used to store and retrieve data such as documents, images, backups, static website files, and application assets. Data is organized inside buckets.

### AWS Lambda

AWS Lambda is a serverless compute service that runs code in response to events without requiring the user to manage servers. It is useful for event-driven processing, APIs, automation, scheduled tasks, and backend functions.

### Service Comparison

| Service | Primary purpose    | Typical use                   |
|---------|--------------------|-------------------------------|
| EC2     | Virtual compute    | Run applications and servers  |
| S3      | Object storage     | Store files and static assets |
| Lambda  | Serverless compute | Run event-driven functions    |

## Conclusion

The Week 1 practice set provided hands-on exposure to AWS identity management, virtual compute, and AWS geographic infrastructure, while establishing a basic understanding of common AWS service models and core services.